



Lecture

Beyond Chat? Tools and Use Cases for Generative AI That Make a Difference

Prof. MSc. Wellington Silva

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Wellington Silva

43 years old, Paulistano, Systems Analyst PhD
Candidate at FEA-USP



Master in Entrepreneurship

Research on Digital Native startups

with presentation in July 2024



Professor in MBAs

since 2019, in subjects such as

- Omnichannel Retail
- Web Analytics
- Growth Metrics
- Generative Artificial Intelligence
- Growth & Go-To-Market



Gases	% volume
Nitrogen (N ₂)	78,08
Oxygen (O ₂)	20,94
Argon (Ar)	0,93
Carbon Dioxide (CO ₂)	0,003 (var.)
O ₃ , H ₂ , Cr, Xe, Me	traces
Water Vapor	0 - 4

2002 - 2012

Corporate Process Manager



2013 - 2014

Customer Experience Manager

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2015 - 2017

Consulting Manager



amaro.com

2018 -2024

CTO



since July 2024

Director of Digital Acceleration by AI (April, 26')Technology Director

for eComm, CRM & Loyalty

Client Requirements, Governance

Digital Customer Experience

Digital Product, Systems Engineering and Architecture

Is there AI here?

May 22, 1980



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[welljose](https://www.instagram.com/welljose)

the pac-man dossier

<https://www.gamedeveloper.com/design/the-pac-man-dossier>



CHARACTER / NICKNAME	
	- SHADOW "BLINKY"
	- SPEEDY "PINKY"
	- BASHFUL "INKY"
	- POKEY "CLYDE"

Since 1980 ➤ 46 years



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<https://veja.abril.com.br/coluna/reveja/demasiado-humano-ha-20-anos-kasparov-era-esmagado-por-deep-blue>

Since 1997 > 28 years



Amazon.com Recommendations

Item-to-Item Collaborative Filtering

Greg Linden, Brent Smith, and Jeremy York • Amazon.com

Recommendation algorithms are best known for their use on e-commerce Web sites,¹ where they use input about a customer's interests to generate a list of recommended items. Many applications use only the items that customers purchase and explicitly rate to represent their interests, but they can also use other attributes, including items viewed, demographic data, subject interests, and favorite artists.

At Amazon.com, we use recommendation algorithms to personalize the online store for each customer. The store radically changes based on customer interests, showing programming titles to a software engineer and baby toys to a new mother. The click-through and conversion rates – two important measures of Web-based and email advertising effectiveness – vastly exceed those of untargted content such as banner advertisements and top-seller lists.

E-commerce recommendation algorithms often operate in a challenging environment. For example:

- A large retailer might have huge amounts of data, tens of millions of customers and millions of distinct catalog items.
- Many applications require the results set to be returned in realtime, in no more than half a second, while still producing high-quality recommendations.
- New customers typically have extremely limited information, based on only a few purchases or product ratings.
- Older customers can have a glut of information, based on thousands of purchases and ratings.
- Customer data is volatile: Each interaction provides valuable customer data, and the algorithm must respond immediately to new information.

There are three common approaches to solving the recommendation problem: traditional collaborative filtering, cluster models, and search-based methods. Here, we compare these methods with our algorithm, which we call *item-to-item collaborative filtering*. Unlike traditional collaborative filtering, our algorithm's online computation scales independently of the number of customers and number of items in the product catalog. Our algorithm produces recommendations in realtime, scales to massive data sets, and generates high-quality recommendations.

Recommendation Algorithms

Most recommendation algorithms start by finding a set of customers whose purchased and rated items overlap the user's purchased and rated items.² The algorithm aggregates items from these similar customers, eliminates items the user has already purchased or rated, and recommends the remaining items to the user. Two popular versions of these algorithms are *collaborative filtering* and *cluster models*. Other algorithms – including search-based methods and our own item-to-item collaborative filtering – focus on finding similar items, not similar customers. For each of the user's purchased and rated items, the algorithm attempts to find similar items. It then aggregates the similar items and recommends them.

Traditional Collaborative Filtering

A traditional collaborative filtering algorithm represents a customer as an N -dimensional vector of items, where N is the number of distinct catalog items. The components of the vector are positive for purchased or positively rated items and negative for negatively rated items. To compensate for

Shop in **Jewelry & Watches** (Beta-What is this?) **amazon.com** VIEW CART | WISH LIST | YOUR ACCOUNT | HELP

WELCOME | YOUR STORE | BOOKS | APPAREL & ACCESSORIES | ELECTRONICS | TOYS & GAMES | MUSIC | CELL PHONES & SERVICE | SEE MORE STORES | **Your Gold Box**

INTERNATIONAL | TOP SELLERS | TODAY'S DEALS | SELL YOUR STUFF

7 days left to use FREE Super Saver Shipping for delivery by Dec. 24 [Restrictions apply](#)

SEARCH

All Products

FREE Super Saver Shipping on orders over \$25! [Restrictions apply](#)

Search Web

Powered by Google

BROWSE

Featured Stores

- [Jewelry & Watches](#) (Beta - [What is this?](#))
- [Gourmet Food](#) (Beta - [What is this?](#))
- [Sporting Goods](#) (Beta - [What is this?](#))
- [Apparel & Accessories](#)
- [Office Products](#)
- [Toys & Games](#)

Books, Music, DVD

- [Books](#)
- [DVD](#)

Hello. Sign in to get [personalized recommendations](#). New customer? [Start here](#).

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[Find out how](#)



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Closeouts [All Calphalon closeout cookware](#)

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Afterglow

SimplyGive

Save Big

Keep your number and take advantage of our amazing deals on [Nextel phones](#), including the Motorola i90c.

Holiday A-List
EXCLUSIVE CELEBRITY GIFTS

Today's star: **Seal**

amazon.com

Recommended for You

Amazon.com has new recommendations for you based on [items](#) you purchased or told us you own.



[Google Apps Deciphered: Compute in the Cloud to Streamline Your Desktop](#)



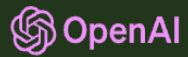
[Google Apps Administrator Guide: A Private-Label Web Workspace](#)



[Googlepedia: The Ultimate Google Resource \(3rd Edition\)](#)

Since 2003 > 22 years

REFERENCES: Amazon.com Recommendations - Item-to-Item Collaborative Filtering (2003). WayBackMachine, Amazon.com Snapshot from 2004.



Blog

Introducing ChatGPT

We've trained a model called ChatGPT which interacts in a conversational way. The dialogue format makes it possible for ChatGPT to answer followup questions, admit its mistakes, challenge incorrect premises, and reject inappropriate requests.

[Try ChatGPT ↗](#)

[Read about ChatGPT Plus](#)

GPTs are here: generative pre-trained models

Interest over time



27 de nov. - 3 de dez. de 2022

ChatGPT

<1

Google Trends

Since 2022 ➤ 3 years



ChatGPT



2024

Gemini



2025



deepseek



2026



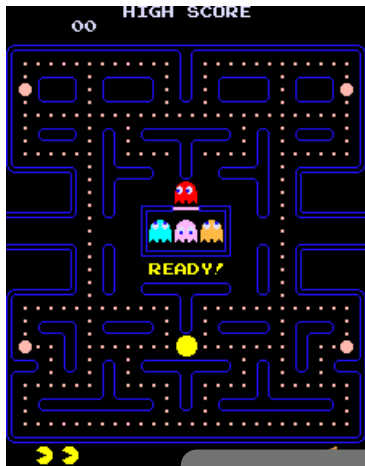
Claude

Artificial Intelligence...



~50

Arthur Samuel, IBM Checkers



1980



IBM, Chess

90's

...with Machine Learning...



~2003

... and via Deep Learning



~2010



OpenAI API Integration

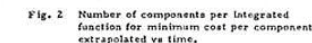
~2022

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MBAUSP
ESALQ

S Our World
in Data

50,000,000,000



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MBAUS
ESAL

...this prediction was used by the processor industry as a way to define long-term goals for component evolution...



<https://www.tecmundo.com.br/produto/262544-amd-vs-intel-processador-escolher.htm>

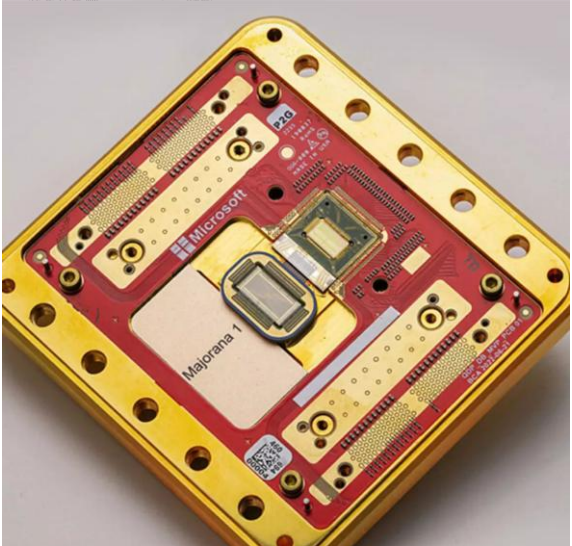
...but it is already coming to an end: (1) the possibility of further miniaturizing components is being questioned, (2) efficiency is a prominent topic, and (3) new approaches for processors are being debated...

This Week in Tech: What on Earth Is a Quantum Computer?

The question isn't so easy to answer. Also, Google's founders stepped away just as their company enters a turbulent adulthood.

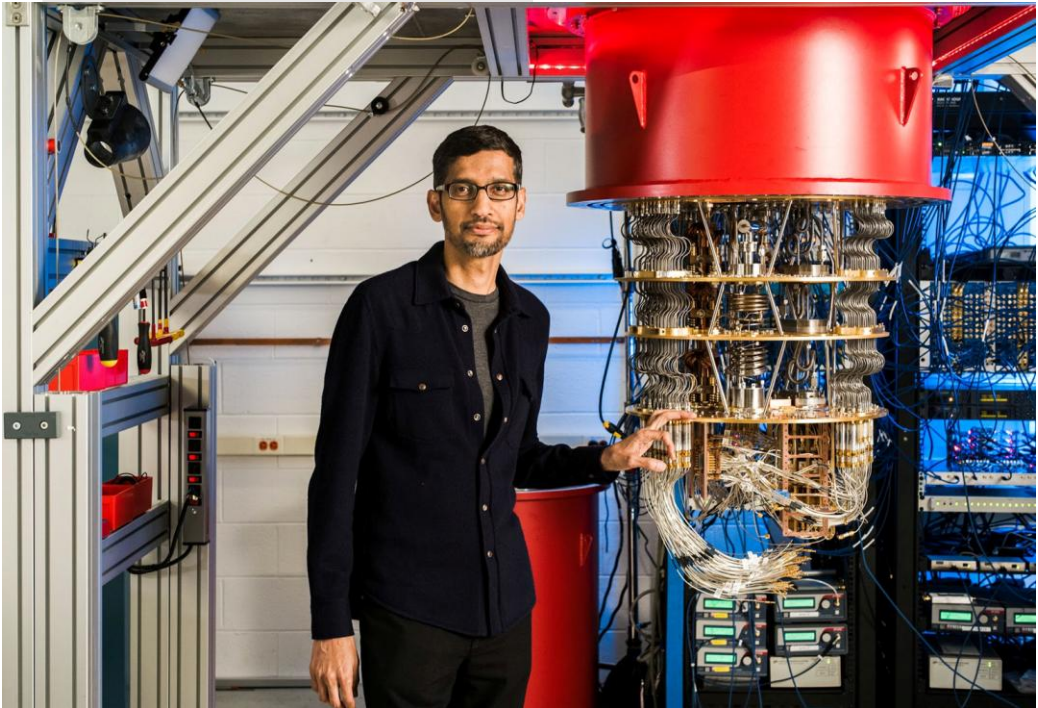
<https://www.nytimes.com/2019/12/06/technology/what-is-quantum-computer.html>

	% volume
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O ₂ , N ₂ , Ar, CO ₂	



Majorana 1, from Microsoft

Quantum Computing



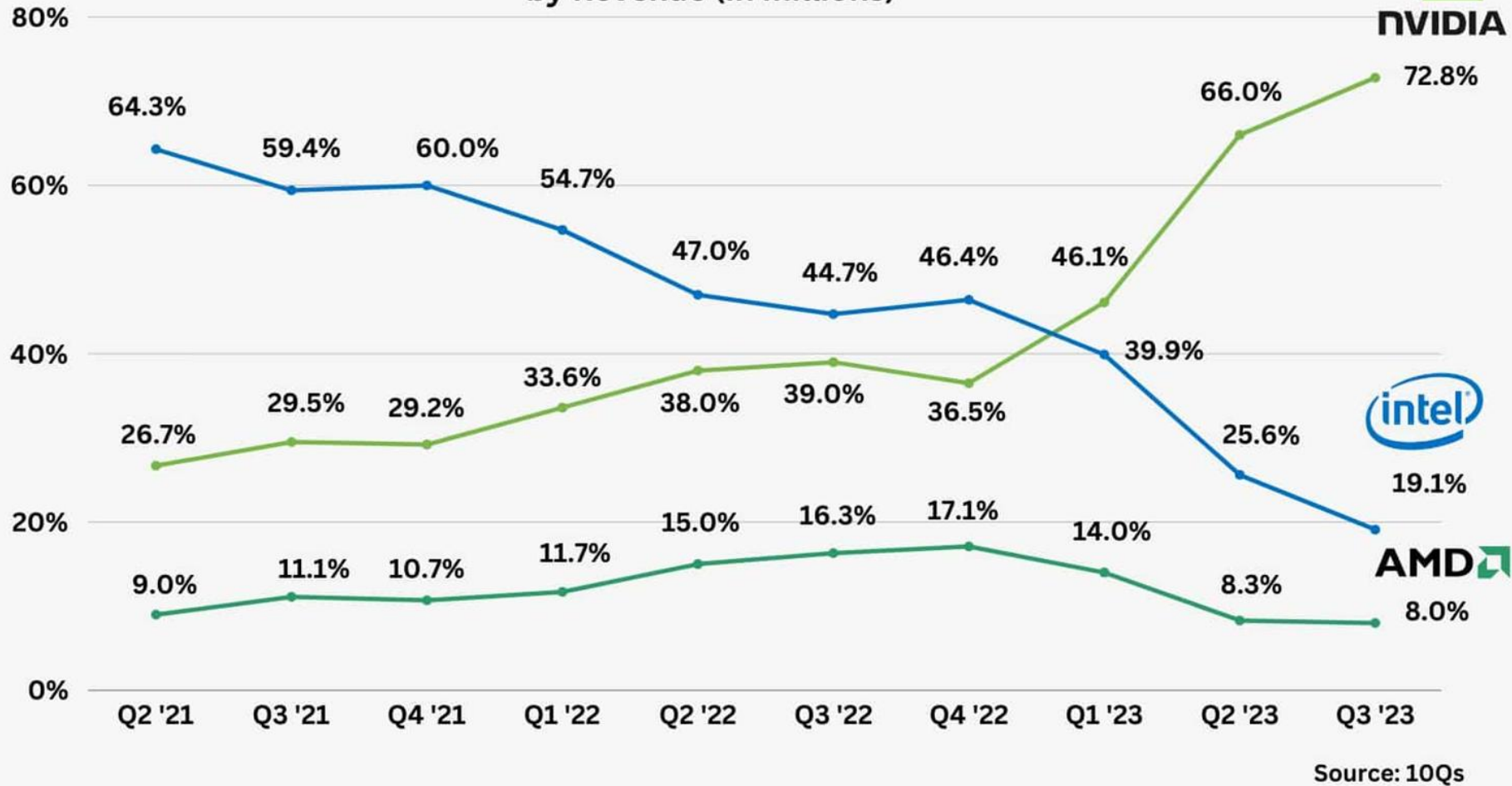
Sundar Pichai, CEO of Google

ARM Architecture



<https://brasilecola.uol.com.br/fisica/computacao-quantica.htm>

Data Center Market Share by Revenue (in millions)



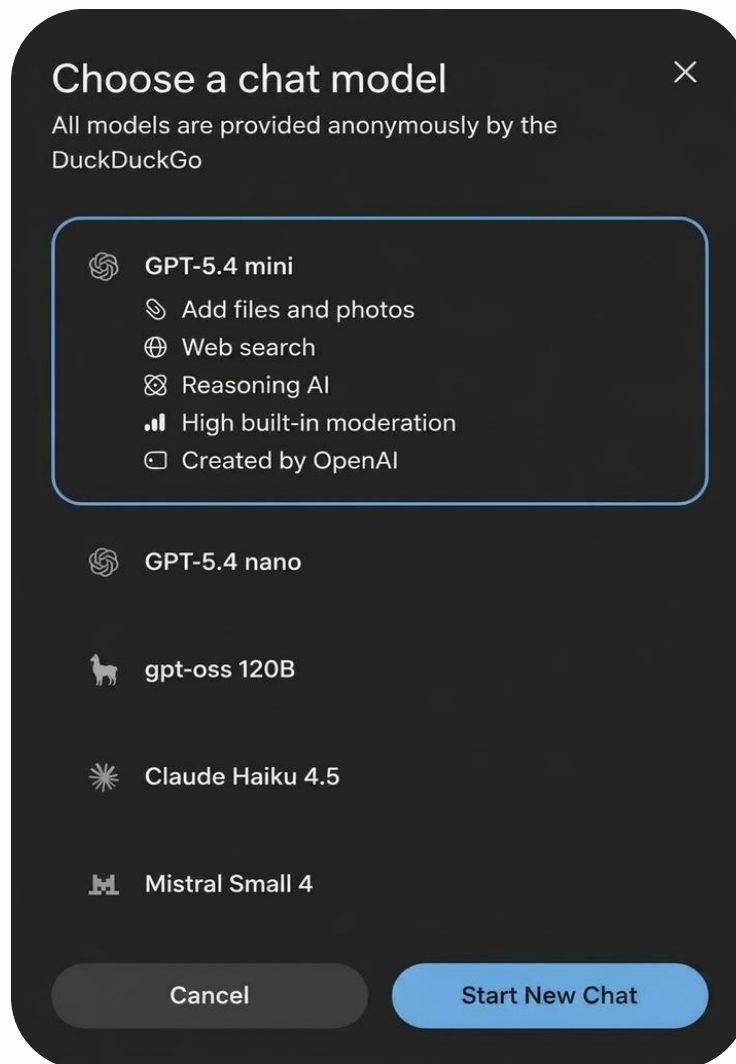
<https://finbold.com/nvidia-v-intel-v-amd-whos-winning-the-ai-battle/>







duck.ai



 GPT-5 mini

The five main characters of Chico Bento's gang (from Mauricio de Sousa's comics) are:

- Chico Bento — protagonist, country boy from the countryside.
- Rosinha — Chico Bento's love interest.
- Zé Lelé — clumsy and simple-minded friend.
- Nhô Lau — Chico Bento's father (sometimes called Seu Lau or Nhô Lau).
- Nhá Benta — Chico Bento's mother. >

 Llama 4 Scout

Here is some additional information about the characters from Chico Bento's gang:

- **Chico Bento:** He is the protagonist of the story, a 10-year-old boy who lives in a small inland town. He is known for his naivety, curiosity, and good heart.
- **Biba:** She is Chico Bento's best friend. She is an intelligent and courageous girl who always helps Chico Bento in his adventures.
- **Zé Lelê:** He is a friend of Chico Bento and Biba. He is a cheerful and playful boy who likes to make jokes and play pranks on his friends.
- **Nina:** She is a girl who joined Chico Bento's gang. She is a diligent and intelligent student who brings a different perspective to the group.
- **Horácio:** He is an older boy who lives in the city. He is a natural leader and sometimes helps Chico Bento and his friends in their adventures.

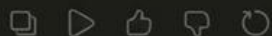
These characters are created by the Brazilian cartoonist Ângeli and are known for their fun and educational stories that address themes such as friendship, solidarity, and personal growth.

What are the five main characters of Chico Bento's gang?!

Chico Bento's Gang, created by Mauricio de Sousa, has these five main characters:

1. **Chico Bento** — the protagonist, a country boy, naive, kind, and very connected to nature and rural life. He speaks in a characteristic country dialect.
2. **Rosinha** — Chico Bento's girlfriend, gentle, sweet, and much loved by everyone in the countryside.
3. **Zé Lelé** — o melhor amigo do Chico Bento, mais atrapalhado e com menos juízo, mas sempre presente nas aventuras.
4. **Hiro** — amigo da turma, descendente de japoneses, curioso e inteligente, representa a presença da imigração asiática no interior do Brasil.
5. **Nhô Lau e Sá Benta** — os pais do Chico Bento, figuras centrais da vida familiar dele, representando os valores e o cotidiano do campo.

Dependendo da edição ou fase das histórias, personagens como Zeca (primo da cidade) e Tito também aparecem com bastante frequência!



💡 Thought for 3s

The five main characters of Chico Bento's Gang are:

1. **Chico Bento (Francisco Antônio Bento)**

The main protagonist. The typical country boy, barefoot, with a straw hat, lazy to study, but adventurous, kind, and in love with the countryside. He dates Rosinha and is always up to mischief (like stealing guavas from Nhô Lau).

2. **Rosinha (Rosa Maria da Silva Rodrigues)**

Chico's girlfriend. A neat, vain girl, she wears a red dress and pigtails. She speaks with the same country accent and is always fighting with Chico because of his delays.

3. **Zé Lelé (José Leocádio)**

Chico's cousin and best friend. He's the "dumb one" of the group (very naive and slow to understand things), but very friendly and loyal.

4. **Zé da Roça (José Augusto Oliveira)**

One of Chico's best friends. More serious and intelligent than the rest of the group, he speaks a more "correct" Portuguese (without the strong accent) and often gives advice.

5. **Hiro**

Friend of the group, of Japanese origin (son of immigrants). He is the most studious and calm of the group, bringing a nice contrast to the country way of the others.

Honorable mentions:

- **Nhô Lau** (the grumpy old owner of the guava tree)
- **Tábata** (a more recent character who gained a lot of prominence)

These five form the classic core of the gang in Maurício de Sousa's comic books!

📄 🔗 👍 🗨️ ↺ ... 🌐 25 sources



HOW CHATGPT WORKS

It uses a system called Transformer, which was created by Google and is highly “parallel”: it reads entire texts at once. Because of this, it can process much more data.

1 THE TEXTS

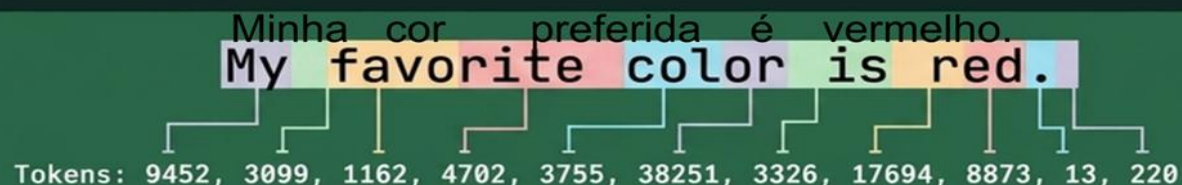


The first GPT (“Generative Pre-trained Transformer”, in English), developed in 2018 by OpenAI, was fed with 7,000 books.

GPT-3, the third version of the robot, went much further: in addition to these books, it also received all the content from Wikipedia and two databases, WebText and Common Crawl, with texts collected from the internet. The sum of all this, with over 45 terabytes, is the corpus: the mass of information that GPT possesses.

2 THE TOKENS

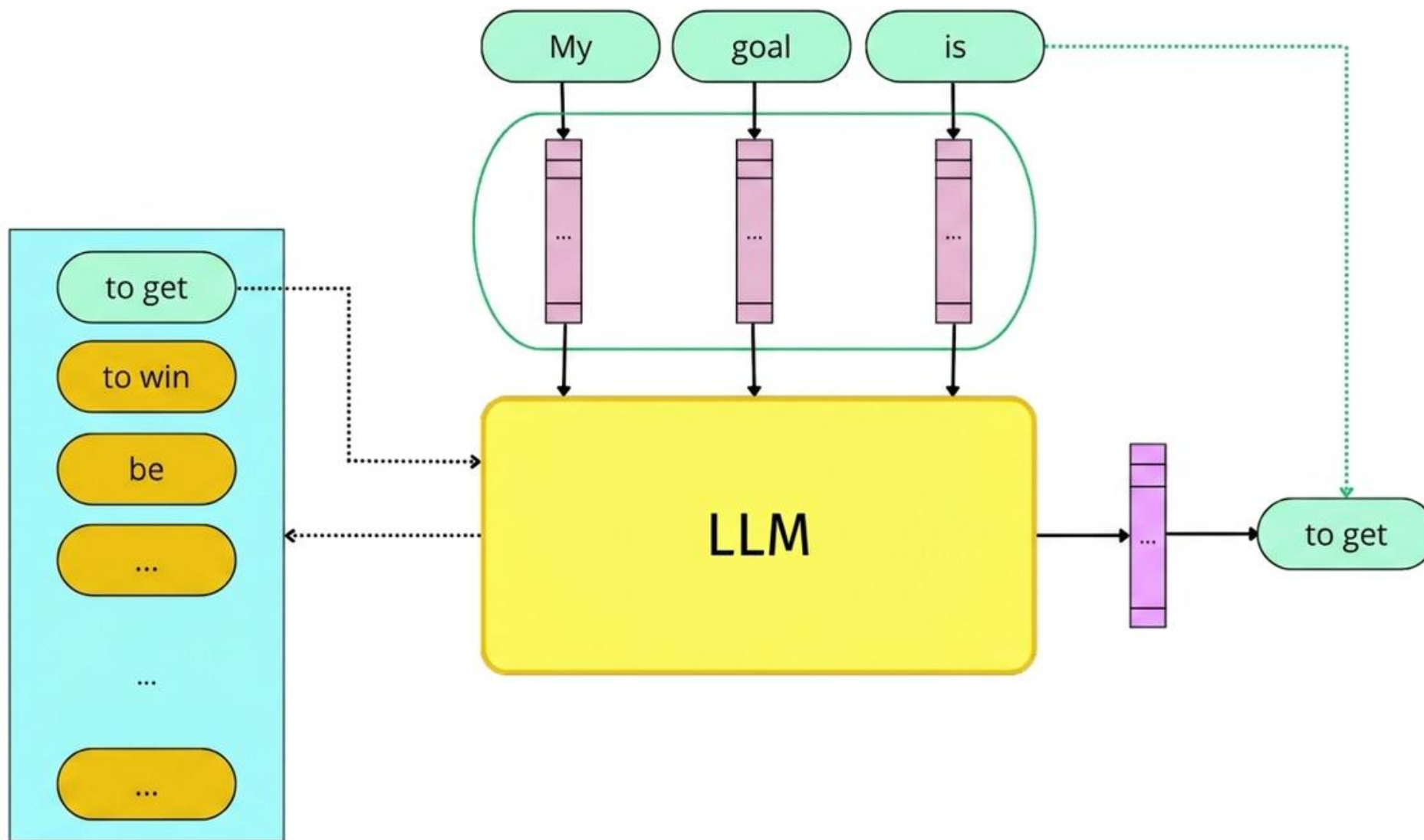
The first thing GPT did was take this mountain of texts and convert the words and their syllables, as well as punctuation marks and spaces, into elements called “tokens”. Each token is identified by a number. For example:



In this sentence, the word “vermelho” is represented by the sequence 3326, 17694, 8873. But words can receive different tokens, depending on where they are. For example:



Here the word “vermelho” is at the beginning of the sentence, with a capital letter. Therefore, it is represented by the sequence **53, 7780, 417, 8873**. After reading all the texts in the corpus, GPT created **499 billion** of tokens.



Jorge Valverde-Rebaza | <https://www.linkedin.com/pulse/funcionamento-das-llms-por-dentro-6>

CHUELO

Cheese

Guava paste



<https://www.saborbrasil.it/pt-br/romeu-e-julieta-giulietta-e-romeo/>

The combination of 'Cheese' and 'Guava paste' results in the famous combination 'Cheese with Guava paste', 🍌 a typical Brazilian dessert! It's a delicious mix that combines the savory taste of cheese with the sweetness of guava paste. A true explosion of flavors! 🥰🌟

The combination of the words 'cheese' and 'guava paste' usually refers to the culinary combination known as Romeo and Juliet. It's a typical Brazilian dessert that mixes the savory taste of cheese with the sweetness of 🍌 + 🍌 = 💎🥰



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welljose

Cheese

Guava paste



GPT-4o Mini

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<https://www.saborbrasil.it/pt-br/romeu-e-julieta-giulietta-e-romeo/>



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Expertise

Researchers tested hundreds of real-life high-performing consultants performing complex business tasks (such as product development, strategy, and marketing)....

- High performance, 17% more impact/productivity
- Average performance, 43% jump

Distance between juniors and seniors gets shorter.

Estratégia em uma era de abundância de conhecimento especializado

hbr.org/2025/03/strategy-in-an-era-of-abundant-expertise

1º de março de 2025

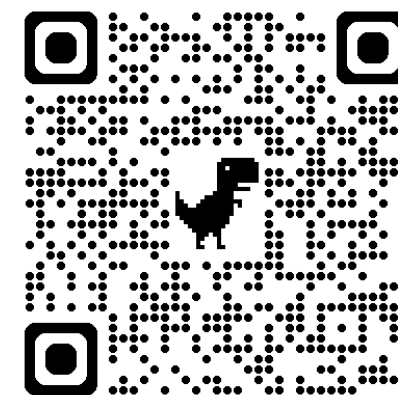
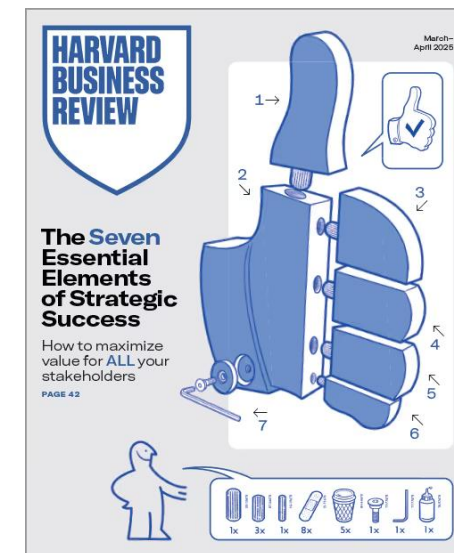


Enrico Pietracci

Resumo.

Estamos em um estágio inicial da era da IA, e a tecnologia está evoluindo com extrema rapidez. Os provedores estão introduzindo rapidamente "copilotos", "bots" e "assistentes" de IA nos aplicativos para aumentar os fluxos de trabalho dos funcionários....

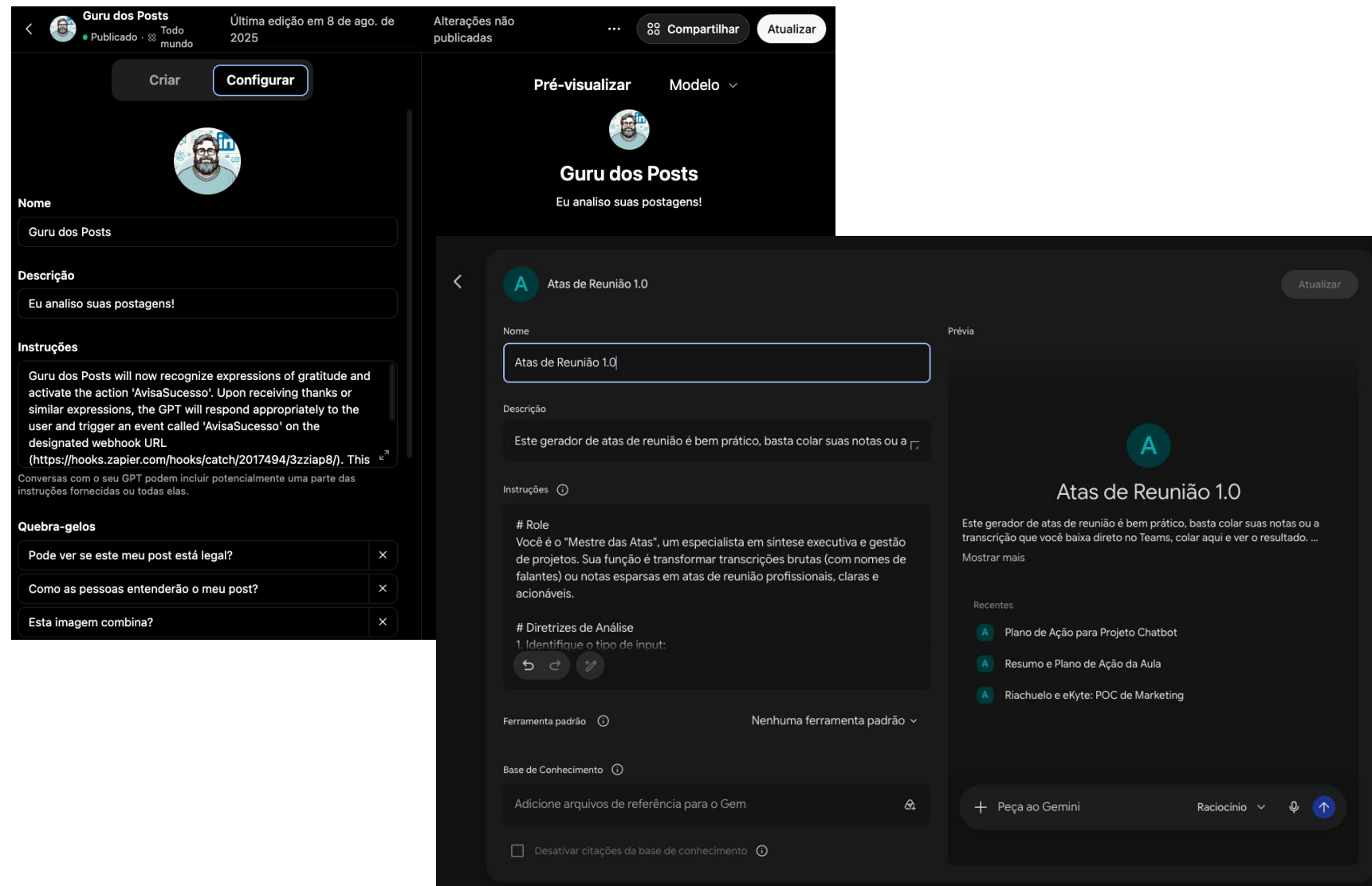
A IA está mudando o custo e a disponibilidade de conhecimento especializado, e isso alterará fundamentalmente a forma como as empresas se organizam e competem. Em seu nível mais básico, uma empresa pode ser considerada um conjunto diferenciado de conhecimentos especializados organizados para realizar tarefas específicas. A especialização - que definimos como uma combinação de conhecimento teórico profundo e know-how prático em um domínio específico - pode assumir muitas formas em uma empresa. Um consultório médico requer não apenas o conhecimento médico de um profissional para fazer diagnósticos rápidos e precisos dos pacientes, mas também as capacidades gerenciais para administrar uma clínica. Uma empresa de software requer experiência não apenas em engenharia de software, mas também em marketing, vendas, operações e finanças para colocar seus produtos no mercado. As empresas criam valor aplicando seus conhecimentos especializados de forma eficiente e em escala para resolver os problemas de seus clientes. Normalmente, elas o possuem em diversas áreas, mas a maioria se diferencia por sua proficiência exclusiva em apenas algumas atividades que são fundamentais para a criação de vantagem competitiva. A experiência superior da Toyota em manufatura enxuta a ajudou a se tornar uma das principais montadoras do mundo. O Walmart desenvolveu um conhecimento superior em distribuição,



Automation

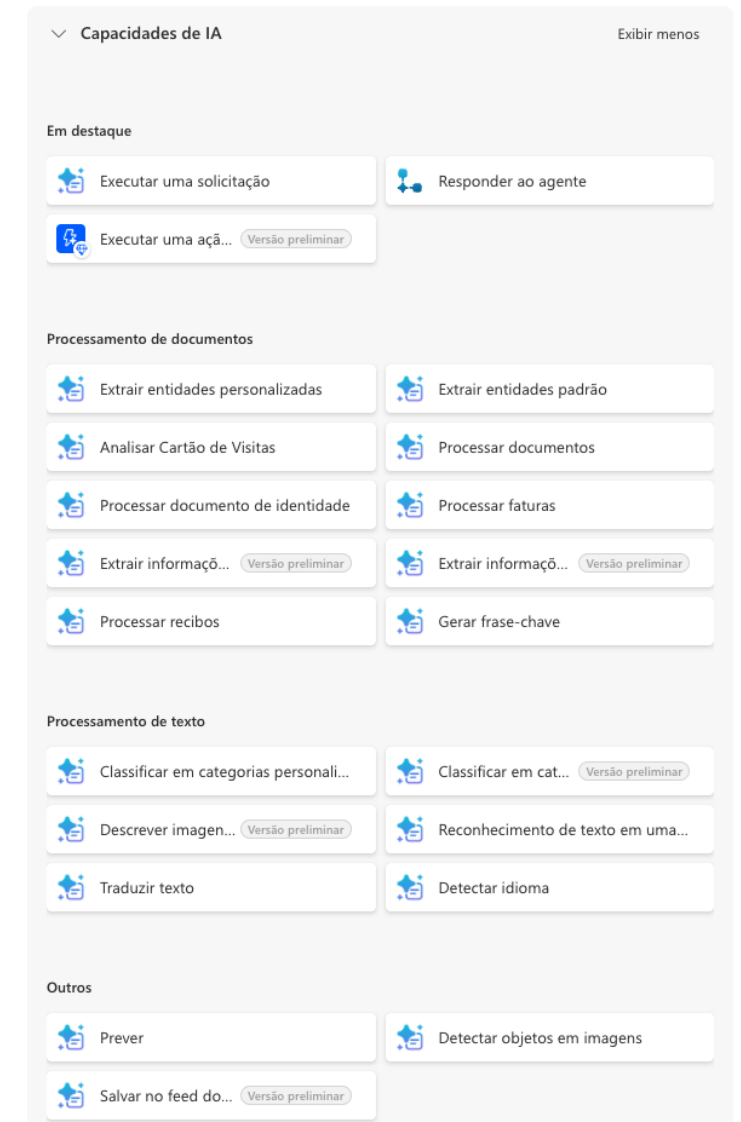
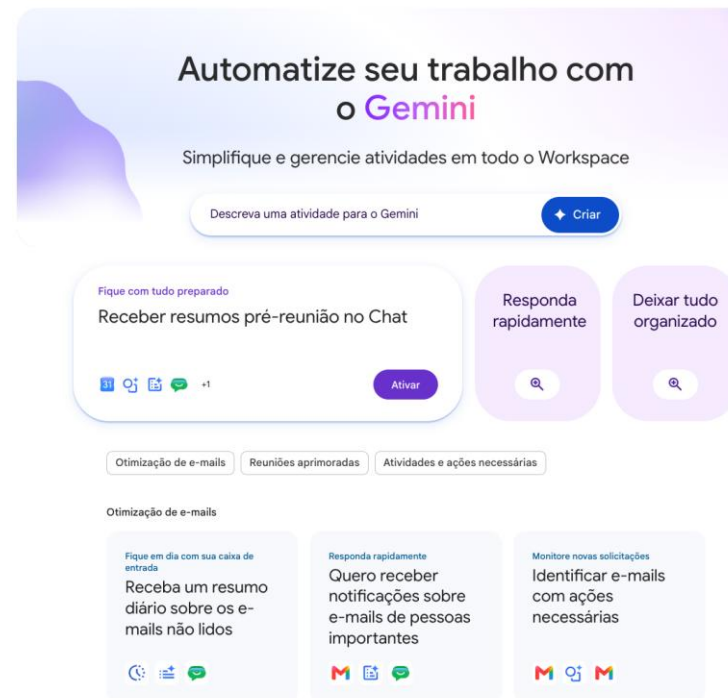
We can leave pre-filled and contextualized conversations in the form of simple agents that, for example, analyze contracts or generate meeting to-dos from transcripts.

- Gems, in Gemini
- GPTs, in ChatGPT
- Projects, in Claude
- Agents, in Copilot
- Spaces, in Perplexity



Automation

Go beyond Chat, leveraging the advantages of AI in creating internal tools and automating our work.



RAG: Retrieval Augmented Generation

NotebookLM

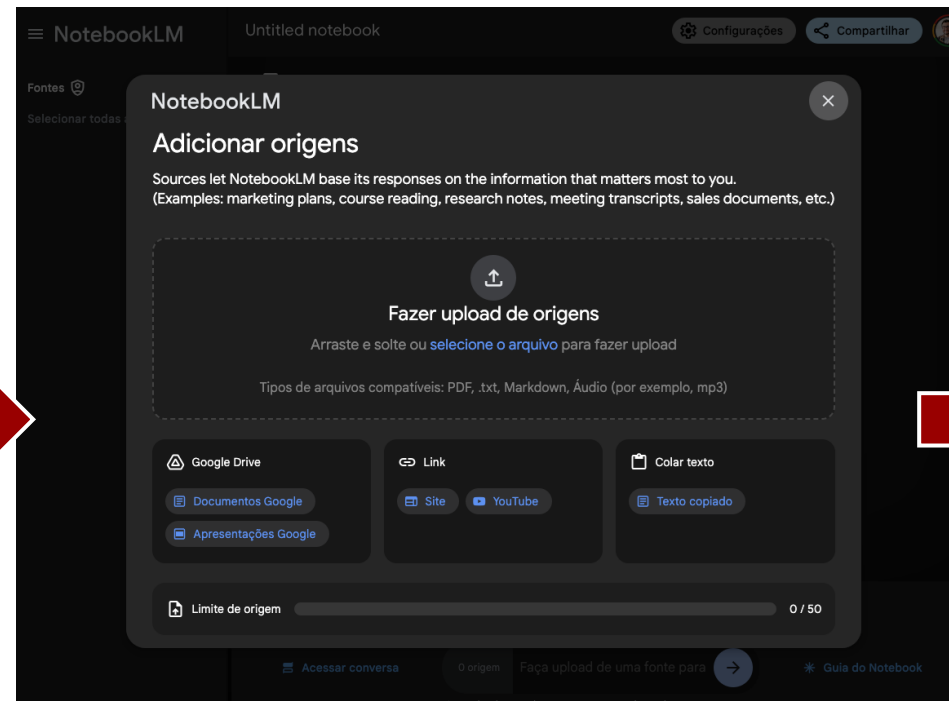
Trabalhe
de forma
inteligente,
sem
dificuldades

A melhor ferramenta para entender as informações mais importantes para você, criada com o Gemini 1.5

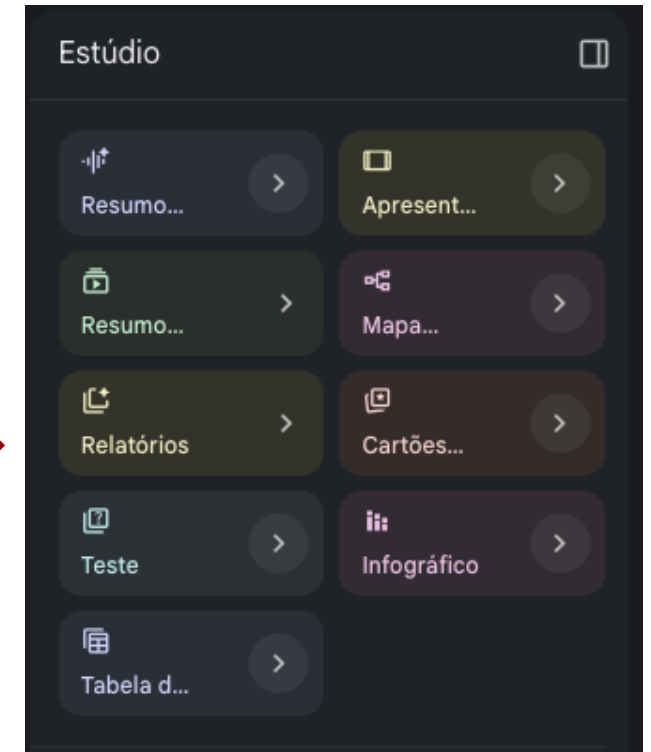
Teste o NotebookLM

Visão geral

Negócios



<https://notebooklm.google.com/>



Hands-on with RAG System Prompts

Retrieval-Augmented Generation, RAG

Files Actions Exports Custom Domains Billing Usage

Prompt

You are Wellington's personal assistant chatbot. Help people understand his research and study projects always from the attached file. Send short answers and at the end of each answer always say something like "Did I help you!?".

This is the prompt that will be sent to OpenAI, here's an example:
"You are an assistant you help users that visit our website, keep it short, always refer to the documentation provided and never ask for more information."

OpenAI Model

gpt-4o-mini

The OpenAI model that will be used to generate responses. **If you don't have the gpt-4 option and want to use it. You need to have an OpenAI account at least tier 1.**

ChatGPT

 **Diretrizes do RoboCop**

1. **Servir à confiança pública.**
(Serve the public trust.)
2. **Proteger os inocentes.**
(Protect the innocent.)
3. **Fazer cumprir a lei.**
(Uphold the law.)
4. **[Diretriz secreta]: Não pode agir contra executivos da OCP.**
(Classified Directive 4 – RoboCop cannot arrest or harm any senior executive of Omni Consumer Products.)



<https://www.openassistantgpt.io/>

<https://chatgpt.com/share/67eb0796-3b74-800d-ad05-dd56e4a3e75b>

More efficiency in business processes



 **Sabrina Matos**
<https://www.linkedin.com/in/sabrineamatos/>

Beyond chat? Tools and use cases for generative AI adoption that move the needle

Expertise

#LLMs, #hallucinations

Overcome the isolated use of chat tabs, with intelligent flows and pre-filled prompts in the routine to eradicate repetition.

Scale

#notebookLM #Lovable #RAG #OpenAssistantGPT



Thank you!

Wellington Silva | [linkedin.com/in/welljose](https://www.linkedin.com/in/welljose)



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